

Cognify – AI-Powered Learning Platform

1. Project Overview

Introduction

Cognify is a full-stack web application designed to help students organize study materials, manage academic tasks, and evaluate learning progress through automated quiz generation and performance analytics.

The platform provides a secure and personalized learning environment where users can create notes, manage tasks, generate quizzes from learning content, and monitor study performance through an interactive dashboard.

2. Project Objectives

The primary objectives of Cognify are:

- Provide a centralized learning workspace for students.
 - Enable secure storage and organization of study materials.
 - Improve productivity through task management.
 - Support self-assessment through automated quiz generation.
 - Visualize learning progress using analytical dashboards.
 - Demonstrate modern full-stack software engineering concepts.
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3. Technology Stack

Frontend Technologies

Technology	Purpose
React 18	Component-based UI development
TypeScript	Type-safe JavaScript
Tailwind CSS	Responsive styling
React Router DOM	Client-side routing
Lucide React	Icon library
Vite	Development server and build tool

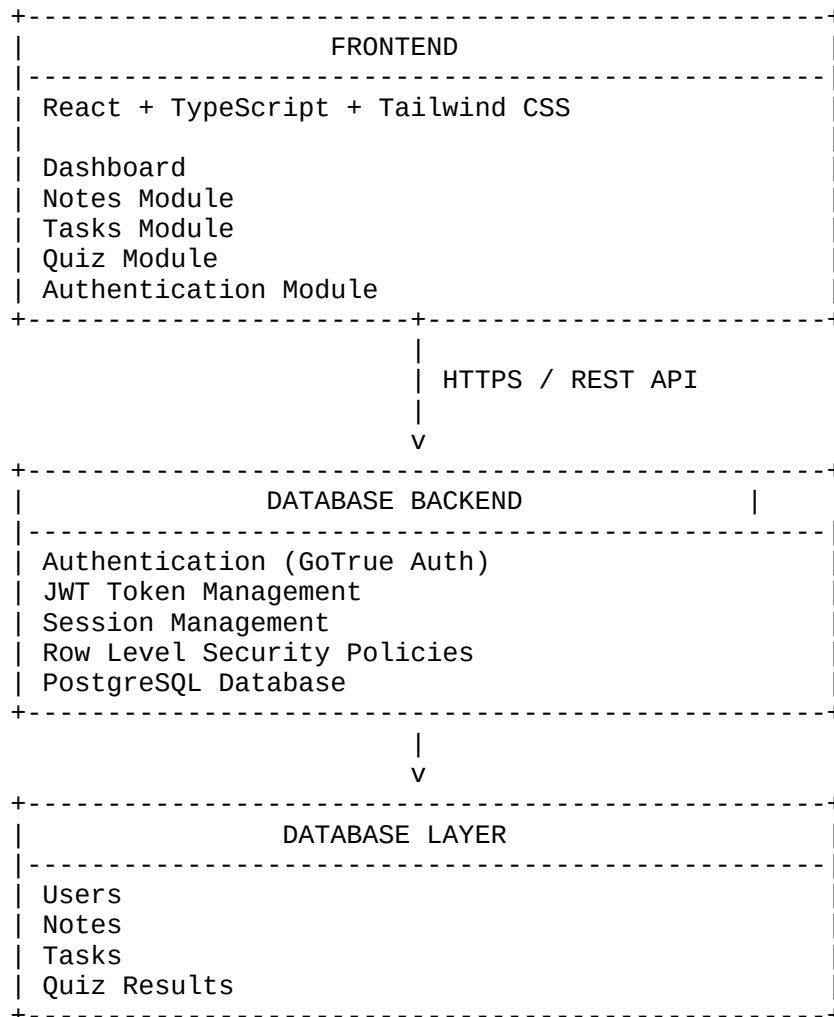
Backend Technologies

Technology	Purpose
Database	Backend-as-a-Service
PostgreSQL	Relational database

Technology	Purpose
GoTrue Auth	Authentication service
Row Level Security (RLS)	Data isolation
JWT	Session authentication

4. System Architecture

High-Level Architecture



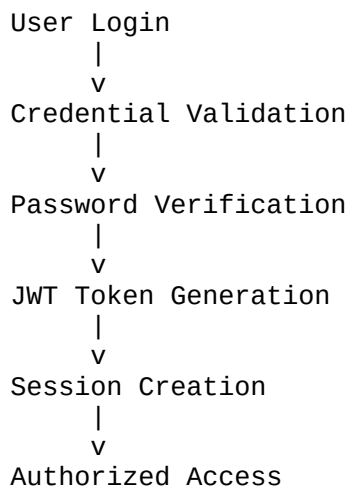
5. Authentication System

Authentication Features

- User Registration
- User Login
- Session Management
- JWT-Based Authentication

- Automatic Token Refresh
- Protected Routes

Authentication Workflow



Security Measures

- Password Hashing using bcrypt
 - Email Validation
 - JWT Authentication
 - Session Management
 - Secure API Access
 - Protected Frontend Routes
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6. Database Design

Database Schema

Users Table

```
id UUID PRIMARY KEY
email TEXT
encrypted_password TEXT
created_at TIMESTAMPTZ
```

Notes Table

```
id UUID PRIMARY KEY
user_id UUID
title TEXT
subject TEXT
content TEXT
created_at TIMESTAMPTZ
updated_at TIMESTAMPTZ
```

Tasks Table

```
id UUID PRIMARY KEY
user_id UUID
title TEXT
description TEXT
deadline TIMESTAMPTZ
completed BOOLEAN
created_at TIMESTAMPTZ
updated_at TIMESTAMPTZ
```

Quiz Results Table

```
id UUID PRIMARY KEY
user_id UUID
note_id UUID
score INTEGER
total_questions INTEGER
created_at TIMESTAMPTZ
```

7. Database Relationships

Users

```
|
+-----< Notes
|
+-----< Tasks
|
+-----< Quiz Results
```

Notes

```
|
+-----< Quiz Results
```

8. Row Level Security (RLS)

Row Level Security ensures users can only access their own data.

Example Policy

```
CREATE POLICY select_own_notes
ON notes
FOR SELECT
TO authenticated
USING (auth.uid() = user_id);
```

Benefits

- Data Isolation
- Multi-Tenant Security
- Database-Level Protection
- Reduced Risk of Unauthorized Access

9. Notes Management Module

Features

- Create Notes
- Edit Notes
- Delete Notes
- View Notes
- Search Notes
- Subject Categorization

Supported Subjects

- Mathematics
- Physics
- Chemistry
- Biology
- History
- Geography
- Literature
- Computer Science
- Economics
- Psychology
- Other

CRUD Operations

Operation	Description
Create	Add new note
Read	View notes
Update	Modify notes
Delete	Remove notes

10. Task Management Module

Features

- Create Tasks
- Edit Tasks
- Delete Tasks
- Mark Complete
- Deadline Tracking
- Progress Monitoring

Task Status Indicators

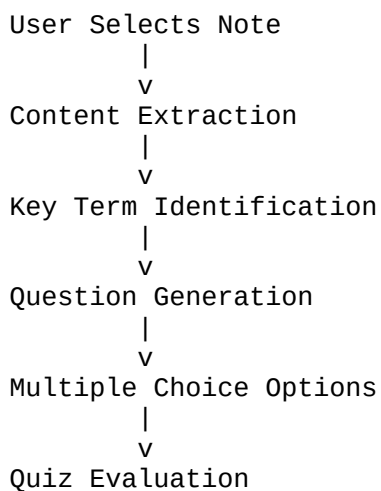
Status	Description
Completed	Green checkmark
Pending	Open task
Overdue	Highlighted warning

11. Quiz Generation System

Purpose

The quiz module automatically generates questions from note content to help students evaluate understanding.

Workflow



Quiz Features

- Automatic Question Generation
- Multiple Choice Questions
- Score Tracking
- Performance History
- Learning Assessment

Scoring System

- Total Questions: 5
- Score Range: 0–5
- Percentage Calculation:

Percentage = (Score / 5) × 100

12. Dashboard Analytics

Dashboard Metrics

Metric	Description
Total Notes	Number of stored notes
Total Tasks	Number of tasks
Completed Tasks	Finished tasks
Pending Tasks	Remaining tasks
Average Quiz Score	Learning performance
Task Completion Rate	Productivity percentage

Dashboard Components

- KPI Cards
 - Progress Bar
 - Recent Notes
 - Upcoming Tasks
 - Quick Action Buttons
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13. User Interface Features

Design Features

- Responsive Layout
- Mobile Sidebar Navigation

- Dark Theme
- Tailwind CSS Styling
- Interactive Components
- Smooth Animations

User Experience Features

- Loading Indicators
 - Error Messages
 - Validation Feedback
 - Progress Tracking
 - Visual Quiz Feedback
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14. File Structure

```
src/
├── App.tsx
├── main.tsx
├── contexts/
│   └── AuthContext.tsx
├── components/
│   ├── Auth/
│   └── Layout/
├── pages/
│   ├── Dashboard.tsx
│   ├── Notes.tsx
│   ├── Tasks.tsx
│   └── Quizzes.tsx
├── types/
├── lib/
└── supabase/
```

15. Software Engineering Concepts Demonstrated

Concept	Implementation
Full Stack Development	React + PostgreSQL
Authentication	JWT
Authorization	RLS Policies
CRUD Operations	Notes & Tasks
Database Design	Relational Schema
State Management	React Context
Type Safety	TypeScript
Responsive Design	Tailwind CSS
Error Handling	Validation & Feedback

Concept	Implementation
Security	JWT + RLS

16. Future Enhancements

AI Features

- Gemini API Integration
- OpenAI Integration
- AI Study Assistant
- Flashcard Generation
- Smart Summaries
- Personalized Recommendations

Advanced Learning Features

- PDF Upload Support
 - Retrieval-Augmented Generation (RAG)
 - Vector Databases
 - AI Chatbot
 - Study Recommendations
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17. Conclusion

Cognify successfully demonstrates modern full-stack software development principles through secure authentication, CRUD operations, database design, analytics dashboards, and responsive user interfaces. The platform provides students with a centralized environment for organizing learning materials, managing academic tasks, and evaluating progress while showcasing industry-standard software engineering practices.